Adaptive partitions for subexponential SAT

A review of twenty ProveIt proposals and smaller explicit contraction bounds

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with ChatGPT

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Abstract

We review the fifteen filed research packets and five newer incoming archives in ProveIt at commit `fd0edd17e6a2`. The newest archives, which are essential to the comparison, lower the best explicit general-degree normalized-wire time coefficient to 35.3545880584 with subexponential space, and to 55.7094783189 with polynomial space. We obtain two further improvements by choosing group partitions from the actual exact-weight mode list sizes. The principal cached construction has time coefficient 34.3252827973 and space coefficient 34.0241780075. An adaptive streamed separator construction has polynomial space and time coefficient 50.2104348364. Each coefficient multiplies $s/\sqrt{\log_2 s}$ in a base-two exponent, with s the normalized Boolean wire budget.

The new finite combinatorial inequalities are proved by exact first and second moments of partitions, and then composed with the existing algebraic contraction interfaces. The full cost analysis pays for recursive group entries, table construction, arithmetic over the actual coefficient ring, and inherited precision. The resulting explicit bound functions are asymptotically smaller than the comparable prior certificates. The logarithmic order of the SAT exponent remains unchanged: this report does not establish an $n/\log n$-type algorithm or a smaller unrestricted $n \log\log n/\sqrt{\log n}$ order. The archive contains mathematical proofs, exact finite verifiers, a nine-factor contraction prototype, source inventories, and an independent review; it contains no complete SAT solver or new proof-assistant formalization.

Dependency boundary. These are refinements relative to the reviewed exact deformation, Hermite recovery, pair-mode separation, packing, and SAT-reduction statements. The upstream preprint is pinned to `openai/math@7ef5c0cc5c`. We do not claim an independent certification of its ETH conclusion.

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1 Results and the comparison standard

The useful comparison starts with the newest incoming material, rather than stopping at the coefficient 76.909 displayed in the project README's round-two ladder. The five later ZIPs improve that frontier substantially [1, 5, 6, 7].

General-degree contraction branch	Time coefficient	Space coefficient
Prior best polynomial-space branch	55.7094783	polynomial
New adaptive streamed separators	50.2104349	polynomial
Prior lower-memory cached branch	37.9301091	33.6230409
Prior fastest cached branch	35.3545881	34.2748030
New adaptive matrix partition	34.3252828	34.0241781

The displayed decimals are descriptive; exact expressions and certified intervals appear below. Space coefficients describe subexponential space, except where the table explicitly says polynomial. The lower-memory prior branch remains a useful option. Low-degree specializations and unproved conditional proposals have different premises and are kept separate.

1.1 The common normalized interface

Let $s \geq 2$, $\lambda = \lceil \log_2 s \rceil$, and let $\mathcal{L}$ be the total graph and tensor-description length. The input is a bipartite Boolean tensor network with at most s wires, polynomially many vertices, and maximum degree at most $\left\lceil s^{9/10} \right\rceil$. Its entries belong to

$$R_0 = (\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^{\ell}), \qquad 1 \leq P_0, \ell \leq s.$$

For fixed constants, each initial entry is computable in $\mathrm{poly}(s + \mathcal{L})2^{O(s^{9/10})}$ time and polynomial space. The constants implicit in these premises are fixed uniformly over the input family. This is the normalization used by the comparison packets [2, 6].

Theorem 1.1 (Adaptive cached contraction)**.** *Under this interface and the inherited exact identities specified in Section 3, the contraction can be computed deterministically in time and space respectively*

$$T(s, \mathcal{L}) \leq (s + \mathcal{L})^{O(1)} 2^{(C_{\mathrm{time}} + o(1))s/\sqrt{\lambda}}, \tag{1}$$

$$S(s, \mathcal{L}) \leq (s + \mathcal{L})^{O(1)} 2^{(C_{\mathrm{space}} + o(1))s/\sqrt{\lambda}}, \tag{2}$$

where

$$\kappa = \frac{\pi\sqrt{40 \log_2 9}}{\ln 2},$$

$$C_{\mathrm{time}} = \kappa\frac{\sqrt{916}}{45} = 34.32528279728494\ldots,$$

$$C_{\mathrm{space}} = \frac{2}{3}\kappa = 34.02417800749051\ldots.$$

In particular, 34.33 *and* 34.03 *are valid eventual coefficients.*

Theorem 1.2 (Adaptive streamed contraction)**.** *Under the same interface, an alternative deterministic construction uses polynomial space and time*

$$(s+\mathcal{L})^{O(1)}2^{(C_{\mathrm{poly}}+o(1))s/\sqrt{\lambda}}, \qquad C_{\mathrm{poly}}=\frac{\pi}{\ln 2}\sqrt{\frac{3625}{42}}\left(1+\frac{1}{3\sqrt{3}}\right)=50.2104348363\ldots.$$

Thus 50.22 *is an eventual polynomial-space coefficient.*

Theorems 1.1 and 1.2 improve the corresponding strongest explicit general-degree certificates found in the twenty packets. They are not claims that every input takes that much time, nor comparisons between measured implementations. For example, writing $f(s)=s/\sqrt{\log_2 s}$, the ratio of the new and old fastest *displayed exponential certificate functions* is

$$2^{-(1.0293052611\ldots+o(1))f(s)}\longrightarrow 0.$$

Polynomial factors contribute $O(\log s)=o(f(s))$ for polynomial-size descriptions. Thus a smaller coefficient here is a strict asymptotic improvement in an explicit upper bound, even though its logarithmic scale remains $f(s)$.

1.2 What remains unresolved

For SAT variables n and clauses m, the strongest established order claimed by the reviewed framework is still

$$\mathrm{poly}_k(n,m)\,2^{O_k\left(\frac{n}{\sqrt{\log n}}\min\{\log(2+m/n),\log\log n\}\right)}. \tag{3}$$

The density-aware packets supply this envelope [8]. The new contraction routines can replace the old engine under the same encodings; they do not change the wire budget that produces the $\log\log n$ factor. The conditional target $2^{O(n\log\log n/\log n)}$ in the precision packet requires a structured summation algorithm that is still missing [9]. We have not established that target or a faster order. No numerical coefficient per SAT variable follows from a coefficient per normalized wire without the encoder's constants and outer-loop accounting.

2 Frozen repository review

The checkout used throughout is `fd0edd17e6a29117179b40a453c0def59165bd7c`. The source and archive identities are recorded in `provenance/source_manifest.json` and `provenance/packet_inventory.json`; a CSV version accompanies them. The review includes the two primary manuscripts, fifteen filed research packets, and all five SAT ZIPs present in `docs/incoming` at that pin. The project README explicitly says that the material was filed as external research and was not checked in ProveIt [1].

Closing placement check. During this review, commit `450252e9d86c` placed the same five incoming packets under `Research/` and restored two previously omitted round-two helper scripts. We compared every archive member with the corresponding new Git blob: all 230 filed files are byte-identical. Thus the closing placement changes the canonical locations, not the twenty-packet mathematical baseline. The exact comparison is recorded in `provenance/closing_placement_check.json`.

Commit	Relevant change
2b925d8800	Places the first seven packets and manuscripts in the new topic project.
3e09608991	Vendors the OpenAI preprint and its source identity.
45233238ae	Summarizes the first-round claimed bounds and conditional targets.
6d3adc4506	Places eight second-round coefficient refinements.
f42479bca2, f898a9fde4, fa6e904d10	Add the five later archives considered here.
450252e9d8	Places those five archives unchanged; all 230 file identities checked.

2.1 All twenty packets

The following is a claims inventory, with close proof inspection focused on the interfaces used by the new theorems and the five newest archives. It is not an assertion that every supplied program or every earlier proof was independently verified. The machine-readable inventory states the review depth and supplies source hashes for every row.

Packet	Contribution and comparison scope
cauchy_mode_improvement	Factorial Cauchy weights with asymptotic slope three; an entropy coefficient improvement, not a new global order.
eth_contraction_improvements	Sharp Hermite budgeting, coefficient 648.44, and streamed packet extraction.
3sat_precision_refinement	Precision near $6MD/u$; exact taper analysis; conditional $n \log\log n / \log n$ target.
sat_density_improvement	Uniform density envelope (3).
eth_density_refinement	Related density gain and precision changes; same unrestricted order.
3sat_refinement	Scalar parity lift and smaller incidence-degree cap; second compression is conditional.
3sat_asymptotic_analysis	Restricted barriers and missing compression/degree interfaces; no stronger unrestricted rate.
eth_asymmetric_packet_balancing	Asymmetric degree bookkeeping and near-lossless padding; general coefficient 76.909.
3sat_round2_refinement	Matching-potential packing, coefficient 84.066, actual-domain rank audit.
eth_balanced_packing	Nearly equal pair loads and coefficient 94.748.
eth_high_moment	Independent higher-moment packing derivation of 94.748.
eth_matching_refinement	Independent matching-potential derivation of 94.748.
eth_packet_refinement	Two-stage labeling, coefficient 95.664.
eth_recursive_refinement	Actual residue-class modes and coefficient 133.207.
eth_joint_cauchy	Full-block slope-four modes; 319.686 polynomial-space and 225.597 cached time certificates.
sat_adaptive_contraction	Adaptive cached partitions and streamed separator sums; polynomial-space 55.7095, cached 36.7503/34.0242.
sat_frontier_improvements	Multiple grouping arities, polynomial-space 63.8574, cached 38.4845, and kernel table work.
subexponential_sat_frontier	Polynomial-space 63.8522, cached 37.9301/33.6230, implementation refinements.
subexponential_sat_kernel_improvements	Exact extraction and adaptive Smith mode profiles; no new uniform logarithmic order.

Packet	Contribution and comparison scope
`subsat_contraction_tree`	Binary table construction and exact fast-matrix bridge; fastest prior 35.3546/34.2748.

The final five names denote the dated ZIPs in the incoming directory. All claims in the table have their stated upstream dependencies. The separate *A $2^{o(n)}$ time algorithm for 3-SAT* manuscript has the same displayed worst-case logarithmic order and no comparable explicit normalized-wire coefficient. Its reported formal development was not supplied in these paths. No numerical superiority over an unspecified constant is asserted.

3 Inherited exact interfaces and the entropy norm

We retain the source contraction algorithm's Cauchy deformation, integral local coordinates, Hermite reconstruction, canonical creation-time representatives, and exact child-network identity [2]. The sharper packing and precision statements are taken from the asymmetric, joint-Cauchy, adaptive, and cached-tree packets [3, 4, 5, 6]. The new work changes how a finite mode sum is evaluated; it does not alter those identities.

At an active recursive call let M be the number of wires, u the number of retained valuation levels, a the packet exponent, and $D = 2^{a-i}$ at valuation level $i < u$. There are r groups. Near-lossless packing gives child size at most $(2/r + o(1))M$. The sharp recovery precision satisfies

$$p_t \le (6 + o(1))MD/u. \tag{4}$$

Full-block Cauchy modes have guaranteed weights

$$w_q = a + 1 - i + 4q - 2s_2(q), \qquad q \ge 0,$$

where $s_2(q)$ is the binary digit sum. For each directed source-target pair there are at most $(2 + o(1))M/(r^2D)$ families of each index q. These facts include the source/target locality needed for factorization.

3.1 Exact budgets and paid group entries

Enumerate the arrays of exact directed-pair weights (b_{gh}) with $\sum b_{gh} < p_t$. The number of entries in an array is the fixed constant $r(r-1)$, so the number of arrays is polynomial in s. For one array let $\mathcal{I}_{gh}$ be the list of nonnegative mode-exponent tuples of exact weight b_{gh}. An empty list makes that array contribute zero. Otherwise the array's contribution is exactly

$$\sum_{(\xi_{gh}) \in \prod_{g \ne h} \mathcal{I}_{gh}} \prod_{g=1}^{r} Z_g((\xi_e)_{e \ni g}). \tag{5}$$

Every entry of Z_g is a real child contraction. Counting, ranking, and unranking a list use the standard integer recurrence

$$D(j, b) = D(j-1, b) + D(j, b - w_j), \quad D(0, 0) = 1,$$

with impossible budgets assigned zero. Its dimensions and integer bit lengths are polynomial in s. Exact list sizes can consequently be used in partition selection before the expensive entries are computed.

Creation-time precision matters here. A packet tensor retains its immutable description and the canonical representatives chosen when it was created. Later higher-precision calls reuse those representatives. A group entry depends only on its incident pair indices; it does not secretly depend on the other groups' indices. This is why caching and conditional separation are valid, and why no free-oracle assumption is being introduced.

3.2 A Euclidean bound for the list logarithms

Write $h_e = \log_2 |\mathcal{I}_e|$ and define

$$K(M,u) = \frac{\pi\sqrt{2}}{\ln 2}\frac{M}{\sqrt{u}}.$$

For a nonnegative vector z on the directed pairs, the inherited weighted-partition estimate is

$$\sum_e z_e h_e \le \left(\frac{\|z\|_2}{r} + o(1)\right) K(M,u). \tag{6}$$

For clarity, its constant follows from

$$\sum_{q\ge 0} -\ln(1 - e^{-\theta(1+4q-2s_2(q))}) = \frac{\pi^2}{24\theta} + O(\log^2(2+1/\theta)).$$

Use parameter θ/z_e for pair e, multiply its logarithmic bound by z_e, and sum. The leading terms in natural logarithms are

$$\theta p_t + \frac{\pi^2}{24\theta}\frac{(2+o(1))M}{r^2 D}\sum_e z_e^2.$$

Optimization and (4) give (6). The retained-level precision buffer makes the remainder negligible uniformly over active depths. The dimension $r(r-1)$ is fixed. Uniformity over the positive unit sphere follows either from the displayed remainder or from a finite epsilon-net followed by a limit. Therefore

$$\|(h_e)\|_2 \le (1+o(1))K(M,u)/r. \tag{7}$$

This norm statement allows the partition to depend on the actual list sizes. Treating all lists as equally large would lose precisely that flexibility.

4 Adaptive nine-group triangle contraction

Let $K > 0$ and let $h_{gh} \ge 0$ be the logarithmic sizes of the exact-weight mode lists, for the 72 directed pairs on nine group vertices. The inherited weighted-entropy estimate implies, uniformly,

$$\|(h_{gh})\|_2 \le (1+o(1))K/9.$$

Here the inference from estimates for each fixed nonnegative weight vector is valid: normalize by K, use the coordinate estimates for boundedness, and apply compactness (or a fixed finite epsilon-net of the positive unit sphere). A sequence violating the asserted uniform norm bound would have a subsequence converging to a vector x of norm greater than $1/9$; the fixed weight vector $x/\|x\|_2$ contradicts the original estimate.

For an unordered pair $e = \{g,h\}$ put $w_e = h_{gh} + h_{hg}$. Write

$$W = \sum_e w_e, \qquad S = \sum_e w_e^2, \qquad d_v = \sum_{e\ni v} w_e.$$

Then $S \le 2\sum_{g\ne h} h_{gh}^2 \le (2+o(1))K^2/81$. A partition of nine vertices into three triples has three interblock classes. Let their weights be H_1, H_2, H_3 and put $H = H_1 + H_2 + H_3$. The exponent of the imported exact rectangular matrix multiplication bridge, using a fixed rational matrix multiplication exponent below $12/5$, is

$$F = H - \frac{3}{5}\min_i H_i.$$

Lemma 4.1. *Among the* 280 *partitions into triples there exists one with*

$$F \le \sqrt{\frac{57}{175}}\,K + o(K).$$

It can be selected with polynomial work from the integer sizes of the exact weight lists, without computing real logarithms.

Proof. Choose the partition uniformly. For each unordered edge e, define $v_e \in \mathbb{R}^3$ to be zero if e is internal, and otherwise to be the standard vector of its interblock class minus $(1/3, 1/3, 1/3)$. A fixed edge is crossing with probability $3/4$. Thus $\mathbb{E}H = 3W/4$ and

$$\sum_i (H_i - H/3)^2 = \left\| \sum_e w_e v_e \right\|_2^2.$$

The covariance inner products are

$$\mathbb{E}(v_e \cdot v_f) = \begin{cases} 1/2, & e = f, \\ 1/28, & e \ne f \text{ share one vertex}, \\ -1/21, & e, f \text{ are disjoint}. \end{cases}$$

For the adjacent case, the two other endpoints lie in the same other block with probability $3/14$ and in distinct other blocks with probability $9/28$. Their dot products are $2/3$ and $-1/3$, respectively, giving $1/28$. For the disjoint case, every partition has nine unordered edges in each interblock class, so $\sum_e v_e = 0$. Each edge has fourteen adjacent edges and twenty-one disjoint edges. The row-sum identity therefore gives $1/2 + 14/28 + 21c = 0$, hence $c = -1/21$. Expanding yields the exact identity

$$\mathbb{E}\sum_i (H_i - H/3)^2 = \frac{8}{21}S + \frac{1}{12}\sum_v d_v^2 - \frac{1}{21}W^2.$$

Let B be the unsigned vertex-edge incidence matrix of K_9. Then $BB^{\mathsf{T}} = 7I + J$. On vectors of zero sum, the largest squared singular value is seven. Applying this to $w' = w - (W/36)\mathbf{1}$ gives

$$\mathbb{E}\sum_i (H_i - H/3)^2 \le \frac{27}{28}\left(S - \frac{W^2}{36}\right).$$

A triple of real numbers of sum zero has maximum at most the square root of two-thirds of its sum of squares. Jensen's inequality consequently gives

$$\begin{aligned} \mathbb{E}F &= \frac{4}{5}\mathbb{E}H + \frac{3}{5}\mathbb{E}\max_i (H/3 - H_i) \\ &\le \frac{3}{5}W + \frac{9}{5\sqrt{14}}\sqrt{S - W^2/36} \\ &\le \sqrt{\frac{4617}{350}}\sqrt{S} \le \left(\sqrt{\frac{57}{175}} + o(1)\right) K. \end{aligned}$$

The penultimate step is two-dimensional Cauchy–Schwarz, since the coefficient of $\sqrt{W^2/36}$ is $18/5$, and $(18/5)^2 + (9/(5\sqrt{14}))^2 = 4617/350$. Some partition is no worse than this mean.

For a partition let $A_i \geq 1$ be the integer product of the list sizes in its ith interblock class. Minimizing F is equivalent to minimizing $(A_1A_2A_3)^5/(\min_i A_i)^3$. Compare these rational quantities by exact cross-multiplication over the 280 partitions. Their bit lengths are polynomial, as in the existing exact-index interface. Zero-sized lists cause the entire exact-budget contribution to be skipped. □

5 Composition with recursive table construction

Use the selected partition. Cache every group factor, paying for each entry with a real recursive child contraction. The incident-pair coefficient is $\alpha = 4/9$. Build each triple table by first merging two singleton tables, then merging with its remaining singleton. For a merge of blocks of sizes a, b, with $k = a + b$, the three matrix index classes contain $2a(9-k), 2ab, 2b(9-k)$ directed pairs. The inherited entropy estimate and the $12/5$ matrix bridge give the coefficient

$$\frac{1}{9}\sqrt{v_1 + v_2 + v_3 - \frac{21}{25}\min_i v_i}.$$

The pair merge has classes $(14, 2, 14)$, with squared numerator $708/25$. The triple merge has classes $(12, 4, 24)$, with squared numerator $916/25$. The triangle lemma is below both relevant dominant costs. Hence the local nonrecursive time coefficient is

$$\beta = \frac{\sqrt{916}}{45} = 0.6725664866854025\ldots.$$

The largest stored boundary has thirty-six directed pairs, so its coefficient is $j = 2/3$. Cache storage has coefficient $4/9$ and is smaller. The paid recurrence is

$$T_d \leq \mathrm{poly}(s)2^{(\alpha+o(1))K_d}T_{d+1} + \mathrm{poly}(s)2^{(\beta+o(1))K_d}.$$

A single polynomial in total input description length pays for entry reading.

Use precisely the already proved fixed-nine-group schedule in the incoming adaptive-contraction packet:

$$\lambda = \lceil \log_2 s \rceil, \quad B_\lambda = \max\{q \geq 0 : 9^{20q} \leq 2^\lambda\}, \quad U = B_\lambda - \lceil \sqrt{\lambda} \rceil, \quad u_d = \lfloor U/2^d \rfloor,$$

with additive buffer $\lceil 6 \log_2 \lambda \rceil$ and leaf threshold $\lceil s/\lambda \rceil$. All packing, precision, canonical creation, entry-oracle, ring-coordinate, and bit-cost gates remain the imported gates; the new computation changes only how the already constructed finite mode sum is evaluated. In particular, the uniform schedule analysis gives

$$K_d \leq (\kappa + o(1))\gamma^d \frac{s}{\sqrt{\lambda}}, \qquad \kappa = \frac{\pi\sqrt{40\log_2 9}}{\ln 2}, \quad \gamma = \frac{2\sqrt{2}}{9}.$$

Because

$$\frac{\alpha}{1-\gamma} = \frac{4}{9-2\sqrt{2}} < \frac{2}{3} < \beta,$$

the root merge term dominates the entire paid recurrence. The resulting relative exact-contraction certificate is

$$C_T = \kappa\frac{\sqrt{916}}{45} = 34.32528279728494\ldots, \qquad C_S = \frac{2}{3}\kappa = 34.02417800749051\ldots.$$

Ancestor tables add instead of multiplying, and the unchanged schedule makes the deeper table exponents decrease geometrically. This is a subexponential space branch, not a polynomial-space theorem. It does not improve the logarithmic order of the SAT exponent or establish a stronger conditional summation interface. It is strictly smaller than the fastest explicit incoming normalized-wire certificate, $35.354588058401\ldots$ time and $34.274803005023\ldots$ space.

6 Arithmetic and schedule details for the cached theorem

6.1 Matrix multiplication over the actual ring

The coefficient ring during recursion is

$$R = (\mathbb{Z}/2^p\mathbb{Z})[U, V]/(f(U), V^\ell),$$

where f is monic of degree $O(\log s)$ and irreducible modulo two. This ring can contain zero divisors and nilpotents; treating it as a field would be invalid. We use the cached-tree packet's integer-coordinate bridge [6]. We spell out its costs because they affect the theorem.

An established rational matrix-multiplication exponent strictly below $12/5$ supplies a fixed finite rational bilinear algorithm [11]. Clear its finitely many constant denominators. The same identities hold over every prime field outside a fixed finite exceptional set. Recursive use of this fixed algorithm gives square cost at most $O(N^{12/5})$ up to fixed constants and quadratic field-element storage.

The ring R is a free module of polynomial rank over $\mathbb{Z}/2^p\mathbb{Z}$, with a polynomial-size integer multiplication table. Lift coordinate inputs to nonnegative integers. A coordinate entry of a product of $d_1 \times d_2$ and $d_2 \times d_3$ matrices is bounded by $d_2(2^p-1)^2$ and has at most $2p + \lceil \log_2 d_2 \rceil + 1$ bits. This is polynomial in s. Polynomially many small nonexceptional odd primes suffice for exact CRT reconstruction. Multiply every required coordinate pair, recover its integer matrix product, apply the structure constants, and reduce modulo 2^p. This adds polynomial factors only.

Blocking rectangular products at side $\min(d_1, d_2, d_3)$ gives bit work

$$\mathrm{poly}(s + \mathcal{L})d_1d_2d_3 \min(d_1, d_2, d_3)^{-3/5}.$$

Input, output, and temporary storage have at most a polynomial factor times their table-cell counts. Thus the logarithmic work objective really is $H_1 + H_2 + H_3 - (3/5)\min H_i$, as used in the adaptive lemma. No nonunit is inverted in R. The prototype in this archive uses ordinary scalar multiplication to test the finite identity; it does not implement the theoretical fast backend.

6.2 Uniform packing and precision

For the fixed-nine schedule in the preceding proof, put $b = \lceil 6\log_2 \lambda \rceil$ and $a_d = \sum_{j=0}^{d} u_j + (d+1)b$. The exact integer definition of U gives

$$U = \frac{\lambda}{20\log_2 9} - \sqrt{\lambda} + O(1), \qquad \sum_{d\ge 0} u_d \le 2U.$$

At the root, with $Q = 9^{a_0}$ and $\Delta = \left\lceil s^{9/10} \right\rceil$, the symmetric high-degree packing term has logarithm at most

$$\log_2(Q^2\Delta) \le \lambda - 2\log_2 9\sqrt{\lambda} + O(\log \lambda).$$

Below the root orient the retained side as the high-degree side. Its degree remains at most Δ, and the fresh side has degree at most $2^{a_{d-1}+1}$. The asymmetric high-degree term $Q\Delta$ has the same negative square-root slack, now with an $O((\log \lambda)^2)$ remainder. The fresh-side and rounding terms $Q^2 2^{a_{d-1}+1}$ and $2^{a_d}Q^2$ have exponents bounded by

$$\frac{2\log_2 9 + 1}{10\log_2 9}\lambda + o(\lambda) < 0.232\lambda + o(\lambda).$$

These gaps dominate all required fixed polynomial logarithmic factors and $s/M < \lambda$ while the call is active.

For a noncircular depth argument, first check hypothetical depths $d \le \lceil \log_4 \lambda \rceil + O(1)$. In that range $u_d \ge \lambda^{1/2+o(1)}$, and the buffer has total size $O((\log \lambda)^2)$. Packing gives child ratio $2/9 + o(1) < 1/4$; this closes the proposed depth bound. The sharper ratio and the taper decay yield the geometric rate $\gamma = 2\sqrt{2}/9$.

At a retained level, $D \ge 2^{a_{d-1}+b+1}$. Inherited precision is $O(s\lambda^2 2^{a_{d-1}})$. Its ratio to MD/u is at most $O(\lambda^4 2^{-b}) = o(1)$. The jet-evaluation loss is also lower order. Thus the precision coefficient six is uniform on active paths. All ring parameters, indices, descriptions, and interpolation lengths remain polynomial in $s + \mathcal{L}$.

6.3 Summing the complete recurrence

Let $\alpha = 4/9$ and $\beta = \sqrt{916}/45$. The local recurrence pays for every child entry before the group tables are combined:

$$T_d \le \mathrm{poly}(s)\left(2^{(\alpha+o(1))K_d}T_{d+1} + 2^{(\beta+o(1))K_d}\right).$$

A path ending at nonrecursive work at depth d has normalized coefficient

$$\kappa\left(\alpha\sum_{j<d}\gamma^j + \beta\gamma^d\right) = \kappa\left(\frac{\alpha}{1-\gamma} + \left(\beta - \frac{\alpha}{1-\gamma}\right)\gamma^d\right) \le \kappa\beta.$$

The strict margin $\alpha/(1-\gamma) < \beta$ makes this stable under the uniform lower-order perturbations. More formally, for any sufficiently small fixed $\eta > 0$, the eventual bound uses $\kappa + \eta$ and $\gamma + \eta < 1$; after summing the geometric majorant, let η tend to zero. This avoids assuming that arbitrary $o(1)$ errors can be multiplied through an increasing number of levels without control.

Polynomial branching factors across $O(\log \lambda)$ depths add $O(\lambda \log \lambda) = o(s/\sqrt{\lambda})$ to the exponent. Direct leaves cost $2^{O(s/\lambda)}$; initial entry evaluation costs $2^{O(s^{9/10})}$; packet entry work is $\exp(O(2^{a_d}\log s))$ with $a_d \le (1/(10\log_2 9) + o(1))\lambda$. All are lower order, including after the paid ancestor branching. One fixed polynomial in the original description length pays for entry access rather than becoming an additional recursive branching factor.

For space, the largest boundary table has 36 directed pairs, so its logarithmic cell count is at most $(2/3 + o(1))K_d$. Ancestor tables add their storage sizes; their exponents do not add. The largest is at the root, giving $C_{\mathrm{space}} = (2/3)\kappa$. Exact ring arithmetic and index records contribute polynomial factors per cell. This completes the resource accounting in Theorem 1.1.

7 An adaptive separator tree in polynomial space

The same second-moment idea gives a separate improvement that retains polynomial space. It changes only the order of the exact group-factor sum; it needs no matrix multiplication. We use the existing streamed separator algorithm and its $r = 16$ precision and packing schedule. Its earlier coefficient is $55.709478318910\ldots$.

Theorem 7.1 (Adaptive streamed coefficient)**.** *Under the same normalized contraction interface and inherited algebraic identities, there is a polynomial-space algorithm with time*

$$(s+\mathcal{L})^{O(1)}2^{(C_{\mathrm{ps}}+o(1))s/\sqrt{\log_2 s}}, \qquad C_{\mathrm{ps}} = \frac{5\pi}{\ln 2}\sqrt{\frac{145}{42}}\left(1+\frac{1}{3\sqrt{3}}\right) = 50.210434836331\ldots.$$

In particular, 50.22 *is a safe eventual coefficient.*

The exact local sum. Fix a pair-budget array for sixteen groups. Each directed pair e has an exact-weight index list of size N_e, and write $h_e = \log_2 N_e$. Empty lists discard the array. The imported entropy bound, in its Euclidean form, gives

$$\sum_e h_e^2 \le (1+o(1))\frac{K^2}{16^2}, \qquad K = \frac{\pi\sqrt{2}}{\ln 2}\frac{M}{\sqrt{u}}.$$

For each unordered pair let $w_{ab} = h_{ab} + h_{ba}$, and put $q = \sum_{a<b} w_{ab}^2$ and $W = \sum_{a<b} w_{ab}$. Thus $q \le (1+o(1))K^2/128$.

Take the full balanced binary tree on sixteen leaves. At a node, condition on all directed indices crossing its two child blocks, recurse independently on the two sides, multiply the answers, and sum. For a leaf ℓ, let E_ℓ be the union of the cut-pair sets on its root-to-leaf path, viewed as unordered pairs, and let

$$L_\ell = \sum_{e\in E_\ell} w_e.$$

The number of paid group-factor evaluations is at most $\sum_\ell 2^{L_\ell} \le 16\,2^{\max_\ell L_\ell}$, as in the inherited streamed separator identity. Each group-factor evaluation is an actual recursive child contraction. No cache or free oracle is used.

Lemma 7.2 (Adaptive tree moment bound)**.** *For every nonnegative weighting of the* 120 *unordered pairs, some relabeling of the balanced sixteen-leaf tree satisfies*

$$\max_\ell L_\ell \le \sqrt{\frac{725}{1344}}\,K + o(K).$$

Proof. Uniformly permute the sixteen group labels over the leaves. A leaf path contains $64+16+4+1 = 85$ unordered cut pairs. Write

$$m = \frac{1}{16}\sum_\ell L_\ell, \qquad V = \frac{1}{16}\sum_\ell (L_\ell - m)^2.$$

Linearity of expectation gives $\mathbb{E}m = (85/120)W = (17/24)W$. For any sixteen real numbers,

$$\max_\ell L_\ell \le m + \sqrt{15}\sqrt{V}.$$

Indeed, if the largest deviation is t, the other fifteen deviations sum to $-t$, so their squared sum is at least $t^2/15$.

The quadratic form $\mathbb{E}V = w^{\mathsf{T}}Bw$ is invariant under permutations of the sixteen vertices. Its entries depend only on whether two unordered edges are equal, share an endpoint, or are disjoint:

$$B_{ee} = \frac{19}{192}, \qquad B_{ef} = \frac{5}{168} \quad (e \sim f), \qquad B_{ef} = -\frac{179}{17472} \quad (e \cap f = \varnothing).$$

Here is a direct derivation of these rational counts. An edge whose least common ancestor has $16, 8, 4, 2$ leaves belongs to respectively $16, 8, 4, 2$ leaf paths; these four edge classes have sizes $64, 32, 16, 8$. Thus the trace of the leaf-centered covariance for a fixed tree is

$$32\frac{1}{4} + 16\frac{3}{16} + 8\frac{7}{64} = \frac{95}{8}.$$

Dividing by 120 gives $19/192$. For any one leaf, the degrees of the sixteen vertices in its path's edge set are $15, 14, 12, 8$, with respective multiplicities $2, 2, 4, 8$. Each vertex's average degree over leaves is $85/8$. The sum of these sixteen squared degree deviations is $495/4$. Subtracting twice the covariance trace leaves the total covariance over ordered adjacent edges, namely $495/4 - 2(95/8) = 100$. Dividing by $16 \cdot 15 \cdot 14 = 3360$ gives $5/168$. Finally, constant edge weights give zero leaf variance. The ordered disjoint-edge covariance sum is therefore $-95/8 - 100 = -895/8$, and its $120\binom{14}{2} = 10920$ terms give $-179/17472$.

For an independent executable check, use the incidence indicators $A_{\ell e} = \mathbf{1}_{e \in E_\ell}$ of one fixed tree, compute

$$\frac{1}{16}\sum_{\ell} A_{\ell e}A_{\ell f} - \left(\frac{1}{16}\sum_{\ell} A_{\ell e}\right)\left(\frac{1}{16}\sum_{\ell} A_{\ell f}\right),$$

and average this value over the 120 equal-edge cases, 1680 unordered adjacent-edge pairs, and 5460 unordered disjoint-edge pairs. Relabeling uniformly is exactly this orbit average. The supplied checker performs all of these counts with rational arithmetic.

For a symbolic verification of the spectral bound, let A be the adjacency matrix of the line graph of K_{16}. Its eigenvalues are 28 on constants, 12 on the fifteen-dimensional vertex component, and -2 on the remaining 104 dimensions. This also follows directly from the unsigned vertex-edge incidence matrix R: $R^{\mathsf{T}}R = A + 2I$ and $RR^{\mathsf{T}} = 14I + J$. Substituting the three rational entries above therefore gives eigenvalues

$$0, \qquad \frac{33}{56}, \qquad \frac{85}{2912}$$

for B on these respective subspaces. Consequently

$$\mathbb{E}V \le \frac{33}{56}\left(q - \frac{W^2}{120}\right).$$

Jensen's inequality and then the two-dimensional Cauchy inequality give

$$\begin{aligned}
\min_{\text{trees}} \max_{\ell} L_\ell &\le \mathbb{E} \max_{\ell} L_\ell \\
&\le \frac{17}{24}W + \sqrt{\frac{495}{56}}\sqrt{q - W^2/120} \\
&\le \sqrt{120(17/24)^2 + 495/56}\sqrt{q} \\
&\le \left(\sqrt{\frac{725}{1344}} + o(1)\right) K.
\end{aligned}$$

This proves the lemma. □

An explicit deterministic choice. An exact subset dynamic program avoids enumerating tree relabelings. Replace each h_e by $b_e = \lceil \log_2 N_e \rceil$, calculated from integer bit lengths, and use unordered weights $w_{ab} = b_{ab} + b_{ba}$. This changes any path weight by less than 170 and hence preserves the coefficient. For every subset S of power-of-two cardinality let $F(S)$ be the minimum possible maximum path weight for a balanced tree on S. For singleton sets put $F(\{v\}) = 0$. If $W(S)$ is the sum of internal unordered weights, then

$$F(S) = \min_{\substack{A \sqcup B = S \\ |A| = |B|}} [W(S) - W(A) - W(B) + \max\{F(A), F(B)\}] .$$

Indeed, the root cut weight is common to all paths, and the two subtree problems optimize independently. Ancestor cuts, when present, are also common additive constants and do not change this optimal substructure. Store a minimizing split and recover a complete minimizing tree.

Compute $W(S)$ first for all 2^{16} masks. The dynamic program has $\binom{16}{1}+\binom{16}{2}+\binom{16}{4}+\binom{16}{8}+1 = 14827$ states. Enumerating only splits with $\min S \in A$ examines $120 + 1820 \cdot 3 + 12870 \cdot 35 + 6435 = 462465$ splits. All arithmetic is on polynomial-bit integers. The selector is therefore deterministic and uses polynomial time and space in s. This exact finite selector does not imply a practical end-to-end SAT speedup. Once selected, the separator sum is streamed depth first. The local recurrence is

$$T_d \le \mathrm{poly}(s)\, 2^{(\beta_{\mathrm{ps}} + o(1))K_d} T_{d+1}, \qquad \beta_{\mathrm{ps}} = \sqrt{725/1344}.$$

All internal-node work is charged to the leaf evaluation count, and the index arrays and accumulators remain polynomial in size.

Global schedule and coefficient. Retain the earlier polynomial-space schedule. Let $\lambda = \lceil \log_2 s \rceil$, $g = \lceil \sqrt{\lambda} \rceil$, $B_0 = \lfloor (\lambda - g)/80 \rfloor$, and

$$u_0 = B_0, \qquad u_d = \lfloor 3B_0/4^d \rfloor \quad (d \ge 1), \qquad a_d = \sum_{j=0}^{d} u_j + (d+1)\lceil 6 \log_2 \lambda \rceil.$$

Use leaf threshold $\lceil s/\lambda \rceil$. The total taper length is at most $2B_0$. The symmetric root gate and the retained-side asymmetric gate have a $\Theta(\sqrt{\lambda})$ logarithmic margin. The fresh-side and padding exponents are at most $9\lambda/40 + o(\lambda) < \lambda$. Near-lossless packing gives sharp child ratio $(1 + o(1))/8$ and eventual ratio at most $1/6$, so active depth is at most $\log_6 \lambda + O(1)$ and $u_d \ge \lambda^{1 - \log_6 4 + o(1)} \to \infty$ there. The existing buffer absorbs inherited precision exactly as in the prior proof. Thus the unchanged schedule gives

$$\sum_d \frac{M_d}{\sqrt{u_d}} \le \left[\sqrt{80} \left(1 + \frac{1}{3\sqrt{3}} \right) + o(1) \right] \frac{s}{\sqrt{\lambda}}.$$

Multiplying by $(\pi\sqrt{2}/\ln 2)\beta_{\mathrm{ps}}$ proves Theorem 7.1. Leaves, tensor entry evaluation, and polynomial factors have the same lower-order bounds as in the inherited schedule. Every recursion is streamed, so space stays polynomial.

The new coefficient is the old $55.709478\ldots$ coefficient multiplied by

$$\frac{\sqrt{725/1344}}{\sqrt{170}/16} = \sqrt{\frac{290}{357}} = 0.90129\ldots.$$

This improves a displayed upper-bound certificate, not a measured running time, and it leaves the SAT exponent's logarithmic order unchanged.

8 SAT consequences and the limits of the improvement

For any fixed encoder and normalized network budget, both new engines give strictly smaller explicit bound functions than their compared predecessors. The cached engine offers the fastest reviewed general-degree coefficient; the streamed engine improves the polynomial-space branch. Existing lower degree or lower memory branches can still be selected when their premises or resource limits apply. A finite portfolio of exact algorithms preserves the minimum applicable time guarantee up to a constant scheduling factor, provided its allowed memory regime is respected.

For bounded clause density, the source uses bounded-degree clause and equality networks with $s = O_k(n+m) = O_{k,D}(n)$. The new engines therefore retain the $2^{O_{k,D}(n/\sqrt{\log n})}$ order. Their general-degree coefficients must not be advertised as numerically beating the much smaller coefficients already proved under special low-degree assumptions.

For arbitrary clause density, the source sparsifies and queries sparse polynomial systems. Its retained monomial encoding has $s = O_k(n \log\log n)$; the density-aware refinement yields the envelope (3). Replacing only the contraction engine preserves these wire counts, and therefore the order. An additional wire-compression or structured-summation theorem is needed for a smaller global exponent.

8.1 Why the simplest order-improvement attempts do not close

The prior reports already explain three relevant limitations [9, 10, 8]. First, a mode inventory with $\Theta(M/D)$ choices of each small weight and budget $\Theta(MD/u)$ can contain $2^{\Omega(M/\sqrt{u})}$ binary exponent vectors even after a total-degree cap is imposed. This is a counting obstruction to those numerical constraints alone, not a SAT lower bound. Second, changing only the bounded Hermite taper still incurs an $\Omega(MD/u)$ multiplier loss at some active level in the stated recovery model. Third, the density-aware sparsification cost model

$$n\left(b\sigma + a\frac{\log(1/\sigma)}{\sqrt{\log n}}\right)$$

is minimized at $\sigma = \Theta(1/\sqrt{\log n})$, leaving the $n \log\log n/\sqrt{\log n}$ order. We do not present these existing barriers as new results.

8.2 An explicit incidence-support certificate

The supplementary note supplies a deterministic illustration of the existing raw-incidence encoding obstruction. For $1 \le q < n/2$, on cyclic indices define quadratic equations over $\mathbb{F}_2$ by

$$P_i(x) = x_i \sum_{j=1}^{q} x_{i+j}, \qquad i \in \mathbb{Z}/n\mathbb{Z}.$$

Their nq quadratic monomials are pairwise distinct across the rows: a collision between two different rows would require two offsets summing to n, impossible when both are at most $q < n/2$. Therefore a nonempty linear combination of r rows has exactly rq monomials; Gaussian row operations alone cannot make the nonzero rows sparser.

In the source incidence lift, the constant coefficient of V is

$$[V^0]F(z,V) = \prod_i \left(1 - z_{ii}\sum_{j=1}^{q} z_{i,i+j}\right).$$

Different rows use disjoint incidence indeterminates. This coefficient has exactly $(q+1)^n$ distinct monomials, all with coefficient 1 or -1; the q^n choices taking one nonconstant term from every row have degree $2n$. They survive modulo 2^{n+1}. Any representation that assigns a distinct pair consisting of a network index and a Boolean wire assignment to every retained incidence monomial must therefore satisfy

$$R2^{s_{\max}} \geq (q+1)^n, \qquad s_{\max} \geq n \log_2(q+1) - \log_2 R.$$

For $q = \lceil \log_2 n \rceil^c$ and the source's $\log_2 R = O(\sqrt{n} \log n)$ degree-array enumeration, this is $(c + o(1))n \log_2 \log_2 n$.

The family satisfies the input premises of the source's sparse-query lemma, including the occurrence threshold without high-incidence branching. We do not claim that the specific family is generated by the source's expander/greedy SAT reduction. The prior analysis already gives a probabilistic 3-CNF-derived obstruction; this explicit example is a supplementary certificate, not a new lower-bound order. Aggregating many incidence monomials, identifying variable copies, or changing the representation can evade it. Those are substantive possible research directions, but no sufficiently cheap general algorithm is supplied here.

9 Verification and reproducibility

The archive separates mathematical dependencies from executed evidence. The asymptotic statements are proved in this report. Finite tests validate the exact combinatorial identities and local implementations used in those proofs; they do not experimentally establish a theorem about all input sizes.

- **Nine-group partition certificate.** The checker enumerates all 280 partitions into triples and verifies all 1,296 entries of the 36-by-36 covariance matrix. It also checks an exact rational positive semidefinite certificate, with no numerical eigenvalue approximation.
- **Full paid schedule.** Rational interval arithmetic certifies

  $$34.325282797284 < C_{\text{time}} < 34.325282797285,$$
  $$34.024178007490 < C_{\text{space}} < 34.024178007491.$$

  Integer inequalities certify strict improvement over the predecessor and closure of the recursion. Forty-one hypothetical depth records at three sample logarithmic sizes check the packing gates; these samples supplement the uniform proof.
- **Exact aggregation prototype.** Thirty-two nine-factor contractions over $(\mathbb{Z}/2^p\mathbb{Z})[V]/(V^2)$, for $p = 1, \ldots, 7$, agree with independent exhaustive edge-index enumeration. Tests cover zero divisors, nilpotent coefficients, singleton dimensions, and unequal list sizes. Another 128 profiles check the exact partition selector against the proved analytic inequality as supplementary numerical tests.
- **Streamed-tree certificate.** A separate exact checker verifies the finite moments and rational matrix inequalities used in the polynomial-space theorem. Its complete counts and constant intervals are recorded in the accompanying result file. An independent checker reconstructs the tree by dyadic least-common-ancestor intervals and verifies all 14,400 ordered covariance entries. The minimax subset selector is checked against all 315 balanced eight-group trees on twelve

weight profiles, and on three sixteen-group profiles. Each sixteen-group run uses exactly 14,827 states and 462,465 splits.

- **Incidence example.** Five exact small cases check row independence, retained support sizes, and coefficient aggregation after variable copies are identified.
- **Independent mathematical review.** Separate derivations checked the adaptive partition inequality, paid recurrences, schedule gates, coefficient-ring bridge, and the scope of the SAT consequences.

9.1 Reproduction

The Python checkers use only the standard library. From the extracted directory run:

```
python3 code/verify_all.py
make pdf
```

The first command regenerates the certificate results and returns a failing exit status on any violated check. The second requires a conventional LaTeX installation with the packages listed in the source. The included PDF was compiled from the included TeX files with shell escape disabled and was rendered for visual review. The bundle contains checksums for its files and the exact pinned source inventory.

The prototype deliberately implements ordinary matrix-shaped summation, so its small examples test correctness of the reorganization. The fast matrix backend is an imported theoretical algorithm. There is no measured end-to-end SAT speedup claim, no Lean or Rocq build, and no claim that the preprint's full proof has been independently certified. Those qualifications also apply to the numerical comparisons in the opening table.

10 Research outcome

The new contribution is the use of exact mode-list size information to choose a partition whose *whole aggregation cost* admits a second-moment bound. In the cached algorithm this reduces the final triangle cost below table construction, exposing a smaller dominant coefficient. In the polynomial-space algorithm it reduces the largest path cost of a streamed separator tree. Both improvements retain the exact finite identities and pay for the original child computations.

This gives genuine strict improvements of explicit exponential certificates in both resource regimes. The next order-level advance must do more: compress the incidence representation, exploit correlations beyond these finite group sums, or provide a different recovery/recursion interface. The present result does not resolve those missing steps.

References

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[11] J. Alman, R. Duan, V. Vassilevska Williams, Y. Xu, Z. Xu, and R. Zhou, *More Asymmetry Yields Faster Matrix Multiplication*, arXiv:2404.16349, SODA 2025. Only the established strict bound $\omega_{\mathbb{Q}} < 12/5$ is used, not a claim about the current record. `https://arxiv.org/abs/2404.16349`.